ChatGPT

# Frequently Asked Questions in MCS-224 (2010–2024)

## AI Basics and Learning Paradigms

**Q: What are Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL), and how do they differ?**
**A:** AI is the broad field of creating systems that exhibit human-like intelligence (reasoning, learning, planning) [1]. Machine Learning is a branch of AI where systems learn patterns from data without explicit programming [2]. Deep Learning is a subset of ML using multi-layer neural networks to automatically learn complex features from large datasets [3]. Key differences: ML requires manual feature extraction and works well on structured data, while DL learns features automatically and excels on unstructured data (images, text) [4] [3]. For example, AI includes rule-based systems and ML, ML includes algorithms like decision trees and clustering [2], and DL uses architectures like CNNs and RNNs for tasks like image and speech recognition [3].

**Q: What are Narrow (Weak) AI, General (Strong) AI, and Super (Artificial Superintelligence) AI? Give examples.**
**A:** Narrow AI refers to systems specialized to one task (e.g. image classification, chatbots). General AI is a theoretical system with human-level broad intelligence across tasks [1]. Super AI (ASI) would surpass human intelligence in all domains (currently speculative). *Example:* Siri or AlphaGo are narrow AI (trained for limited domains) [1]. No true General or Super AI exists yet; they remain goals in AI research.

**Q: What is an intelligent agent? What are its properties?**
**A:** An *agent* is an entity that perceives its environment through sensors and acts upon it through actuators to achieve goals. Key properties include:
- **Reactivity:** Responds to changes in the environment.
- **Proactiveness:** Takes initiative (goal-directed behavior).
- **Autonomy:** Operates without human intervention.
- **Adaptability:** Learns from experience to improve.

For example, a **Simple Reflex Agent** acts only on current percepts, whereas a **Model-based Reflex Agent** maintains an internal state to handle partial observability and makes decisions based on history. The model-based agent is more flexible but requires more resources.

**Q: Compare supervised, unsupervised, and reinforcement learning.**
**A:** These are the three main ML paradigms:
- **Supervised Learning:** Learns a mapping from inputs to outputs using labeled data (e.g., classifying emails as spam/not-spam). Algorithms: decision trees, support vector machines, logistic regression [5].
- **Unsupervised Learning:** Finds structure/patterns in unlabeled data (e.g., clustering customers by behavior). Algorithms: K-means clustering, hierarchical clustering.
- **Reinforcement Learning:** Learns by interacting with an environment to maximize cumulative reward (e.g., game-playing agents). Key algorithms include Q-learning, SARSA, and policy gradient methods.

**Q: What are descriptive, predictive, and prescriptive analytics?**
**A:** These are analytics types used in data analysis:
- **Descriptive Analytics:** Summarizes historical data to reveal past patterns/trends (answers "what happened?"). *Example:* Charting last month's sales to detect seasonality [6].
- **Predictive Analytics:** Uses data/models to forecast future outcomes (answers "what might happen?") [7]. *Example:* Using past browsing data to predict products a user will buy.
- **Prescriptive Analytics:** Recommends actions to achieve optimal outcomes (answers "what should we do?") [8]. *Example:* Optimizing delivery routes by combining traffic data with predictive models.

## Search and Game-Tree Algorithms

**Q: Explain state-space search. What factors affect its complexity?**
**A:** State-space search treats a problem as a graph of *states* (nodes) and *actions/transitions* (edges) [9] [10]. To solve, one starts from the *initial state* and applies actions to reach a *goal state*. Key factors:
- **State definition:** Determine state variables and how actions change them [10].
- **Initial/Goal States:** Clearly define starting configuration and desired goal.
- **Action (Transition) Set:** Specify possible moves from each state.
- **Branching Factor:** Average number of successors per state – higher values make the search tree wider [11].
- **Depth:** Steps from initial state to goal – deeper trees increase search time [12].
- **Completeness and Optimality:** Whether a search is guaranteed to find a solution or the best solution given costs [13].

**Q: Describe Depth-First Search (DFS) and Breadth-First Search (BFS), including their time/space complexity and pros/cons.**
**A:**
- **DFS:** Explores as deep as possible along each branch before backtracking. *Complexity:* Time $O(V+E)$ and space $O(V)$ (where $V$=vertices) [14] [15] (due to recursion stack). *Advantage:* Low memory (only stack), can find a solution without exploring all states. *Disadvantage:* Not guaranteed shortest path, can get stuck in deep paths or loops.
- **BFS:** Explores all nodes at one depth level before next. *Complexity:* Time $O(V+E)$ and space $O(V)$ [16] [17] (queue of frontier). *Advantage:* Finds shortest path in unweighted graphs; complete. *Disadvantage:* High memory use (stores frontier) and can be slow if depth is large.

**Q: What is Best-First Search vs. Greedy Best-First Search?**
**A:** *Best-First Search* uses a heuristic $f(n)$ to pick the most promising node (often $f=g+h$ as in A*). Greedy Best-First Search uses only the heuristic $h(n)$ (estimate to goal) to decide which node to expand [18]. Greedy BFS is faster (only considers goal distance) but not optimal; Best-First (like A*) considers cost so can be optimal if heuristics are admissible.*

**Q: Explain adversarial search and the Minimax algorithm.**
**A:** Adversarial search is used in two-player competitive games (zero-sum). It models the game as a tree of alternating moves by two players with opposite goals [19]. Unlike single-agent search (BFS/DFS) used for puzzles, adversarial search assumes opponents make optimal moves.
- **Minimax Algorithm:** A recursive search through the game tree where one player tries to maximize a utility and the opponent minimizes it [20]. It picks the move that maximizes the minimum payoff (anticipating opponent's best response).
- *Properties:* Complete on finite trees, but exponential time $O(b^d)$ in branching $b$ and depth $d$.
- *Alpha-Beta Pruning:* Optimization that skips branches that cannot affect the final decision, reducing effective branching.

**Q: What are A *and AO* algorithms?**
**A:** *A* search is an optimal best-first search for single-agent problems that uses $f(n) = g(n) + h(n)$ (path cost plus heuristic) and guarantees optimality if $h$ is admissible. *AO* (And/Or) search is used for problems with subgoals and uncertainty, combining "AND" nodes (all children needed) and "OR" nodes (choose one) in an And/Or graph; it generalizes A* to handle problem decomposition.

**Q: Compare Iterative Deepening DFS (IDDFS) and DFS.**
**A:** IDDFS performs DFS to increasing depth limits (depth 0,1,2,...). *Complexity:* Time is still exponential but repeated work makes it more costly than DFS alone (roughly $O(b^d)$), space is $O(d)$ (similar to DFS) [14] [15]. *Advantage:* Combines DFS's low memory with BFS's completeness and optimality (finds shortest solution without huge memory). *Disadvantage:* Some overhead of revisiting states multiple times at different depths.

## Logic and Knowledge Representation

**Q: Define propositional logic vs predicate logic.**
**A:** *Propositional logic* uses Boolean variables (propositions) with connectives; it cannot express relations or quantification. *Predicate logic* (first-order logic) extends propositional logic with predicates over objects, variables and quantifiers (∀, ∃). It can express statements about objects (e.g. "All humans are mortal" as $\forall x \, \mathrm{Human}(x) \to \mathrm{Mortal}(x)$). Predicate logic is more expressive, but inference is more complex.

**Q: What are DNF, CNF, and Prenex Normal Form (PNF)? Transform $\neg(A \to (\neg B \land C))$ to DNF.**
**A:** *Disjunctive Normal Form (DNF)* is an OR of AND-clauses. *Conjunctive Normal Form (CNF)* is an AND of OR-clauses. *Prenex Normal Form* has all quantifiers at the front of a first-order formula. To convert $\neg(A \to (\neg B \land C))$ to DNF:
- First remove implication: $A \to (\neg B \land C)$ is $\neg A \lor (\neg B \land C)$.
- Negate: $\neg[\neg A \lor (\neg B \land C)]$ becomes $A \land \neg(\neg B \land C)$.
- Distribute negation: $A \land (B \lor \neg C)$.
- Distribute to DNF: $(A \land B) \lor (A \land \neg C)$.

**Q: What is Skolemization? Why use it?**
**A:** *Skolemization* removes existential quantifiers by introducing Skolem functions or constants. Given a formula in prenex form, each ∃ variable is replaced by a new function of surrounding universally quantified variables [21]. This preserves satisfiability and is used to convert formulas to a form suitable for resolution (CNF) [22]. It simplifies automated theorem proving by eliminating ∃, making formulas easier to handle.

**Q: Explain resolution and unification in logic.**
**A:** *Resolution* is an inference rule: from clauses $C \lor A$ and $\neg A \lor D$, infer $C \lor D$. It's the basis for automated theorem proving in propositional and first-order logic. In AI, resolution is used in expert systems, NLP, and theorem provers to derive conclusions [23].
*Unification* is the process of making two logical expressions identical by finding substitutions for variables [24]. For example, unifying $Loves(x, \mathrm{Mary})$ and $Loves(\mathrm{John}, y)$ yields $x = John, y = Mary$ [24]. In AI, unification is essential in resolution and logic programming (e.g., Prolog) to match hypotheses with facts [24]. Together, resolution with unification enables powerful deductive reasoning.

**Q: What is a knowledge representation frame, script, or semantic net?**
**A:** These are structured ways to represent knowledge:
- **Frame:** A data structure for stereotyped situations, like an object with attributes (slots) and values. E.g.

a “house” frame might have slots for number of rooms, color, etc. Frames often support inheritance and defaults.
- **Script:** An event sequence structure (like a frame for stereotypical events). E.g. a restaurant script with roles: Customer, Waiter, sequence of actions (enter, order, eat, pay).
- **Semantic Network:** A graph of nodes (concepts/objects) and links (relations). E.g. nodes for “Dog”, “Animal”, link “Dog–is-a–Animal”. Good for inheritance and relational data.

**Q: What are forward chaining and backward chaining?**
**A:** These are inference strategies in rule-based systems:
- **Forward chaining (data-driven):** Start with known facts and apply rules to infer new facts until a goal is reached. *Example:* Given facts and production rules, repeatedly apply rules whose premises are met to derive conclusions.
- **Backward chaining (goal-driven):** Start with a query (goal) and work backwards, finding rules that conclude the goal, and recursively prove their premises. *Example:* Prolog uses backward chaining, trying to prove a goal by matching rules.

**Q: What is the Relevance of resolution and unification in AI?**
**A:** Resolution combined with unification is a complete inference mechanism for first-order logic. *Relevance:* It allows AI systems to derive new knowledge by logical deduction. For example, resolution is used in automated theorem provers, expert systems (e.g. medical diagnosis), and NLP for semantic reasoning [23]. Unification is crucial for matching variables and applying rules. Together, they enable machines to handle symbolic reasoning and deductive inference systematically.

## Machine Learning and Data Mining

**Q: Supervised vs Unsupervised Learning – key differences and algorithms?**
**A:** *Supervised learning* uses labeled data to train a model that maps inputs to outputs. *Algorithms:* Regression (linear, logistic), Decision Trees, SVM, Neural Networks. *Unsupervised learning* finds structure in unlabeled data. *Algorithms:* K-means clustering, Hierarchical clustering, PCA. Supervised learning can predict specific targets; unsupervised explores data patterns.

**Q: What is the bias-variance tradeoff? How do we use regularization?**
**A:** *Bias* is error due to wrong model assumptions (underfitting); *variance* is error due to model complexity (overfitting). Total error = bias$^2$ + variance + irreducible error. Regularization (L1, L2 penalties) adds a cost for large weights to reduce variance and prevent overfitting, thus achieving a better bias-variance balance.

**Q: Compare regression vs classification.**
**A:** In *regression*, the target variable is continuous (predicting a numeric value, e.g., house prices) [25]. In *classification*, the target is categorical (predicting labels, e.g., spam or not) [25]. Both are supervised learning. For regression, common loss functions include MSE and MAE; for classification, metrics include accuracy, precision, recall etc (see below).

**Q: Differentiate linear, multiple linear, logistic, and polynomial regression.**
**A:** All model output as a function of inputs:
- **Linear Regression:** $y = w_0 + w_1 x$ (straight line fit).
- **Multiple Linear:** Like linear but with multiple inputs $y = w_0 + w_1 x_1 + ... + w_n x_n$.
- **Polynomial Regression:** Fits a polynomial $y = w_0 + w_1 x + w_2 x^2 + ...$ to capture non-linear trends.
- **Logistic Regression:** Despite the name, it's a classifier; it models $P(y = 1|x)$ with a sigmoid of a linear combination. It predicts probabilities and is used for binary classification.

**Q: What are Precision, Recall (Sensitivity), Specificity, and Accuracy?** (use confusion matrix)
**A:** Definitions from the confusion matrix: [26]
- **Accuracy:** $\frac{TP+TN}{TP+TN+FP+FN}$ – proportion of correct predictions [26].
- **Precision (Positive Predictive Value):** $\frac{TP}{TP+FP}$ – fraction of predicted positives that are true positives [27].
- **Recall (Sensitivity):** $\frac{TP}{TP+FN}$ – fraction of actual positives correctly identified [26].
- **Specificity:** $\frac{TN}{TN+FP}$ – fraction of actual negatives correctly identified [26].

**Q: Supervised vs Unsupervised vs Reinforcement Learning (again).**
**A:** (See above in ML paradigms.)

**Q: Describe Naive Bayes Classifier.**
**A:** A *Naive Bayes* classifier applies Bayes' theorem with the "naive" assumption that features are conditionally independent given the class [28]. It computes $P(y|x) \propto P(y)\prod_i P(x_i|y)$. Key points:
- It is simple and efficient, with few parameters [28].
- Assumes each feature contributes independently to the class (often not true in reality, hence "naive") [28].
- Used for high-dimensional data like text (e.g., spam detection).

**Q: State Bayes' theorem. Interpret its terms.**
**A:** Bayes' theorem (in classification) is $P(H|E) = \frac{P(E|H)\,P(H)}{P(E)}$, where: [29] [30]
- $P(H)$ = Prior probability of hypothesis $H$.
- $P(E|H)$ = Likelihood of evidence $E$ given $H$.
- $P(E)$ = Marginal likelihood (evidence).
- $P(H|E)$ = Posterior probability of $H$ after seeing $E$.
It allows updating beliefs; e.g. computing disease probability given a positive test.

**Q: Give an example illustrating Naive Bayes classification.**
**A:** *Example:* Predict if **"PlayGolf"** = Yes/No given weather features. Compute $P(Yes|X)$ and $P(No|X)$ using prior and likelihoods from data (as in a weather dataset [31]). Choose class with higher posterior. (Details omitted due to space; see standard ML texts.)

**Q: What is Ensemble Learning? Describe one ensemble method.**
**A:** *Ensemble learning* combines multiple models to improve accuracy [32]. It relies on the idea that a group of weak learners can make better predictions together. *Example:* **Random Forest** (a bagging ensemble) builds many decision trees on random subsets of data/features, then averages (regression) or votes (classification) [33]. This reduces overfitting and often yields high accuracy.

# Clustering and Unsupervised Methods

**Q: Clustering vs Classification – differences and examples.**
**A:** *Classification* is supervised: data are labeled and we train a model to assign labels. *Clustering* is unsupervised: algorithm groups similar data points without preassigned labels. *Example (classification):* Identifying animals in photos (labels given). *Example (clustering):* Grouping customers by purchasing behavior without pre-set categories.

**Q: Compare K-means clustering with hierarchical clustering.**
**A:** *K-means:* Partitions data into $K$ clusters by minimizing within-cluster variance. It requires choosing $K$ and is efficient for large datasets (assigns points to nearest centroid, then recompute centroids).

*Hierarchical clustering:* Builds a dendrogram (tree) of clusters by iteratively merging or splitting (agglomerative or divisive). No need to pre-specify number of clusters. It can capture nested cluster structure but is more computationally intensive for large data.

**Q: What is PCA (Principal Component Analysis)?**
**A:** PCA is a dimensionality reduction technique that finds orthogonal directions (principal components) capturing most variance in data. It projects data onto these components to reduce dimensionality while preserving variance. Useful for visualization and pre-processing.

**Q: Describe K-Nearest Neighbors (KNN) classification with an example.**
**A:** KNN classifies a new point by looking at the labels of its $K$ nearest neighbors in feature space. *Example:* Given 2D points with labels, for $K = 3$ and a query at (10,7), find the 3 closest training points and take a majority vote on their class. (Details: compute distances and majority vote.)

## Optimization and Reinforcement Learning

**Q: What is the Turing Test and the Chinese Room argument?**
**A:** *Turing Test:* A test of machine intelligence where a machine is considered intelligent if a human judge cannot reliably distinguish it from a human through dialogue.
*Chinese Room (Searle):* A critique: a person who doesn't understand Chinese follows rules to reply in Chinese and convinces others they understand, showing syntax (symbol manipulation) alone doesn't imply true understanding or "thinking".

**Q: Explain model-free vs model-based reinforcement learning.**
**A:** *Model-based RL:* Learns or uses a model of the environment (state transition probabilities) and uses it to plan (e.g. via value iteration).
*Model-free RL:* Learns policies or value functions directly from experience (e.g. Q-learning, SARSA) without explicitly modeling transitions.
**Subclasses:** For model-free, *policy-based* (learn policy directly) vs *value-based* (learn value function, e.g. Q-values).

## Summary

The questions above cover the most frequently asked topics in MCS-224 exams (2010–2024). Each answer begins with an introduction and uses bullet points for clarity. Citations to relevant AI/ML literature are provided for definitions and key facts [1] [33]. For questions requiring diagrams (none explicitly asked for diagrams above), one would include the relevant diagram as requested.

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